Brain Stroke Classification Using Proposed Segmentation-Guided Confidence Weighted Ensemble Approach

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***Abstract*—Stroke is a time-sensitive neurological condition that requires timely and precise diagnosis to prevent death and long-term disability. In this paper, we examine the automatic classification of brain strokes from CT (computerized tomography) images by identifying the type of stroke as either normal, ischemic or bleeding using ten different state-of-the-art deep learning architectures. While the classification does provide the type of stroke, it does not provide any spatial information regarding the area of the brain that has been affected, so we also have included segmentation of the stroke lesions to provide information on where the stroke lesions are located and delineated, which improves both the clinical interpretability and performance of the model. To accomplish this, we have developed a new segmentation-guided classification strategy, in which we utilize lesion probability maps, generated by the best performing segmentation model, as spatial priors to guide the feature extraction to clinically relevant areas through a learned attention gate. Finally, the predictions of the five different architecture-based classifiers would be fused using our proposed Confidence Weighted Ensemble, which optimally combines the per-model fusion weights and the temperature parameters to increase the robustness and generalizability of the ensemble. Using a publicly available dataset of brain CT images, we have determined that the ensemble produced a Macro F1-score of 99.04% and an AUC of 99.96%, while the Swin-UNet produced the best segmentation results with a Dice score of 0.7533 and an IoU score of 0.6497.**

***Index Terms*—Brain Stroke Detection, CT Imaging, Deep Learning, Convolutional Neural Networks, Medical Image Segmentation, Segmentation-Guided Classification, Ensemble Learning, Explainable AI, U-Net, ConvNeXt, Swin-UNet, Grad-CAM++**

I. Introduction

Stroke is still one of the leading causes of death and long-term disability around the world, and is therefore, placing a greater strain on health care systems. Stroke occurs when blood flow to the brain is interrupted leading to rapid cell death of brain cells and loss of brain function. Stroke is typically defined clinically as either Ischemic (caused by occlusion of blood vessels) or Bleeding (caused by rupture of blood vessels), and must be accurately diagnosed quickly in order for the patient to receive proper treatment. An incorrect diagnosis could lead to poor outcomes for the patient.

Computed tomography (CT) imaging has become the standard modality for assessing acute stroke due to its availability and rapidity [1]; therefore, the research community has placed a significant amount of effort into developing automated methods for analyzing CT images using deep learning [2]. For example, Kuo et al. demonstrated high sensitivity for detecting bleeding stroke on non-contrast CT scans using a convolutional neural network architecture called ResNet [4]; however, this convolutional neural network architecture did not perform as well in detecting small lesions. Further, Phong and Duong utilized DenseNet [5] to classify multi-class strokes better than previous models due to its use of feature reuse and improved gradient flow. Lee et al. evaluated the impact of compound scaling on the performance of EfficientNet [7]; they reported that, when training data are limited, an architecture that uses compound scaling of the depth, width, and resolution of an image yields excellent performance. MobileNetV2 [9] demonstrated an option for deploying deep architectures that consume fewer computational resources. Inception-V3 [10] and Xception [11] confirmed that multi-scale feature extraction with depthwise separable convolutions has value in a wide range of tasks. Finally, ConvNeXt [12] began to combine the advantages of both convolutional networks and transformer-like designs by incorporating large-kernel equations and modern design principles.

For segmentation tasks, U-Net [13] established the encoder–decoder paradigm with skip connections as a standard in medical imaging. Extensions such as Attention U-Net [15], employed by Salehi et al., improved lesion localization by suppressing irrelevant background regions, while ResUNet

[14] incorporated residual learning to enhance feature propagation. More recently, transformer-based approaches have been introduced to capture long-range dependencies. Yuan et al. utilized a Swin Transformer [16] encoder for ischemic stroke segmentation, achieving improved boundary delineation compared to conventional CNN-based approaches.

Despite these advances, most prior work treats classification and segmentation as independent tasks, limiting clinical applicability. Classification models provide diagnostic labels without spatial localization, whereas segmentation models generate lesion masks which is a segmented region without explicit diagnostic categorization. Given these limitations, integrating both tasks into a unified, interpretable framework with ensemble-level robustness remains an open challenge.

In order to address these limitations, this work proposes a unified pipeline that integrates ten classification models and four segmentation models, combined through structured model selection, segmentation-guided fine-tuning, confidence-weighted ensemble fusion, and Grad-CAM++ [17] explainability. The primary contributions of this work are as follows:

- The development of a unified multi-stage framework integrating stroke classification, lesion segmentation, and explainable AI for comprehensive CT-based stroke analysis.
- A systematic best model selection process has been employed that ensures architectural diversity across the various pipeline stages.
- A **segmentation-guided classification** mechanism in which a learned spatial attention gate, conditioned on Swin-UNet lesion probability maps, focuses feature extraction on clinically relevant stroke regions.
- A **confidence weighted ensemble** that jointly optimizes per-model fusion weights and temperature parameters across five architecturally diverse classifiers, improving robustness over any individual model.

II. Methodology

The proposed framework investigates a detailed deep learning approach that incorporates multiple stages. CT scan images are sent through initial preprocessing stages, then passed through ten independently trained classification models and four independently trained segmentation models. Once the machine learning models have been trained, a model selection phase separates out the best performing classifying model for each of five model architectures, and the resulting classifying models are further refined through the use of the segmentation attention from the highest-performing segmenting model. The resulting classifier model predictions will be combined into a **Confidence Weighted Ensemble Model**. Finally, Grad-CAM++ is applied to visualize and quantify the spatial basis of predictions.

The dataset was collected from **two different publicly available sources on Kaggle** [20], [21] to ensure diversity in CT scan imaging samples and stroke patterns. The dataset consists of a total of **8123 brain CT images**, grouped into **three diagnostic classes**: **Normal (3123 images)**, **Ischemic Stroke (2500 images)**, and **Bleeding Stroke (2500 images)**. For the Ischemic and Bleeding classes, **pixel-level segmentation masks are provided**, resulting in **2500 ground truth masks for each class**, enabling supervised training of the segmentation models.

The system architecture is illustrated in Fig. 1, which provides an overview of the brain stroke detection system.

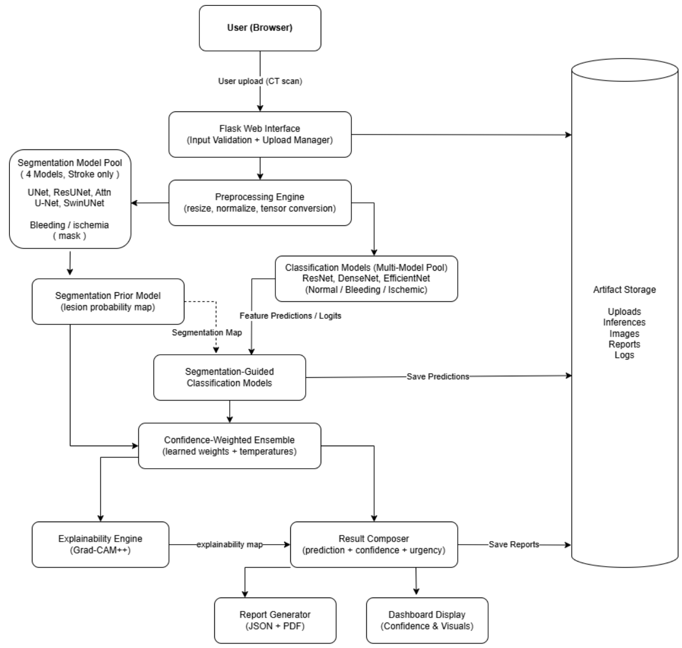

Fig. 1. System Architecture Diagram

A. Preprocessing

All CT images undergo a standardized preprocessing pipeline before being passed to any model, consisting of the following stages.

1) CT Windowing: The CT scans used in this project will be windowed using the recommended window of a center of 40 HU and a width of 80 HU; this is an accepted clinical brain stroke window [1]. Thus, the recommended windows will help improve the separation of the intensity of the stroke lesions against the intensity of the tissue surrounding the stroke.

2) Resizing and Normalization: The images are resized to 224×224 pixels for classification models and 256×256 pixels for segmentation models. Pixel intensities are normalized using ImageNet mean and standard deviation to ensure compatibility with pretrained weights.

3) Data Augmentation: Only training data are augmented to increase variability and support generalization. Augmentation strategies differ between tasks.

Classification: Random horizontal flipping, affine transformations (rotation, translation, scaling), CLAHE [3] (clip limit 2.0, tile grid 8×8, probability 0.6), random gamma correction, brightness and contrast variation, Gaussian blur, Gaussian noise, and coarse dropout.

Segmentation: All classification augmentations are included, with the addition of elastic transformations to strengthen the variability in anatomical structures. Each of the different augmentations applied to the input image and the lesion mask will be used consistently.

At inference time, only resizing and normalization are applied.

B. Classification Models

All 10 classification models were pre-trained from the ImageNet database, and then they were fine-tuned using transfer learning on the stroke dataset. All models had their respective 3-class final classification heads replaced.

Training configuration:

- **Optimizer:** AdamW **Weight decay:** 1×10^{-4}
- **Learning rate:** 1×10^{-4}
- **Batch size:** 32 **Epochs:** 30

1) Baseline and Comparative Models: There are 9 baseline comparison models having different architecture design. ResNet50 [4] utilizes 50 layers with residual skip connections, allowing for stable gradient flow during the extraction of stroke-specific deep feature representations while avoiding degradation of those features due to intensive computational processing. ResNet101 [4] extends the same architecture as ResNet50; it has 101 layers and is competent at capturing subtle differences in textures between ischemic and bleeding lesions, although it requires additional computing power compared to ResNet50. DenseNet121-SE [5] has the same densely connected feature reuse design across 121 layers with an added Squeeze-and-Excitation (SE) channel attention module [6], which allows for improved amplification of hypodense regions that define early-stage ischemic strokes when there is feature reuse between adjacent spatial locations. DenseNet201-SE [5] has an architecture similar to DenseNet121-SE, but allows for generating comparatively rich multi-scale representations of stroke boundaries through the addition of the same SE channel attention module and additional connectivity from layer to layer.

EfficientNet-B4 with Spatial Pyramid Pooling applies compound scaling [7] across depth, width, and resolution, and the SPP module aggregates multi-scale context that helps distinguish small bleeding foci from surrounding tissue. EfficientNetV2-S [8] uses fused-MBConv blocks for improved training speed while retaining the compound scaling benefit, making it effective for capturing both fine and coarse stroke patterns. MobileNetV2 employs inverted residual blocks with linear bottlenecks [9] and depthwise separable convolutions, providing a computationally efficient alternative that achieves competitive classification performance while remaining deployable on resource-constrained hardware. InceptionV3 applies parallel convolutions at multiple kernel sizes within each Inception module [10], capturing stroke features at different spatial scales simultaneously, which is advantageous given the wide variation in lesion size across patients. Xception extends this idea by replacing the Inception modules with depthwise separable convolutions throughout [11], allowing more efficient learning of spatial and channel-wise correlations relevant to CT stroke patterns.

2) ConvNeXt-Small: ConvNeXt-Small is selected as one of the 10 classification models because of its excellent ability to capture both local and global context information. It is an updated version of the CNN architecture that utilizes several vision transformer inspired elements such as depthwise convolutions 7×7 with large kernels, inverted bottlenecks, and layer normalization. It also maintains the efficiency of computations that are achieved by CNNs. For stroke CT scan analysis, these characteristics will assist in determining where stroke lesions exist, as well as discarding normal tissue based on relatively small changes in intensity along large areas of the lesion boundaries.

The core transformation in ConvNeXt blocks can be represented as:

$$Y = X + \mathrm{Conv}_{1\times1}\left(\sigma\left(\mathrm{Conv}_{7\times7}(X)\right)\right) \tag{1}$$

In Equation (1),

- X: input feature map
- $\mathrm{Conv}_{7\times7}$: depthwise convolution capturing spatial context
- $\sigma(\cdot)$: non-linear activation (GELU)
- $\mathrm{Conv}_{1\times1}$: pointwise convolution for channel mixing
- Y: output feature map after residual addition

Additionally, ConvNeXt employs layer normalization instead of batch normalization:

$$\hat{x} = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \tag{2}$$

In Equation (2),

- $\hat{x}$: normalized output at a given layer.
- x: input value to the normalization layer.
- μ: mean of the input computed over the current layer.
- σ^2: variance of the input computed over the current layer.
- ϵ: small constant added for numerical stability.
- γ, β: learnable scale and shift parameters.

Due to the ability of ConvNeXt-Small to provide large receptive fields [12], improved normalization, and a streamlined architectural design, it achieves superior results in the classification of medical images, particularly in the identification of subtle variations in intensity that may be associated with stroke.

C. Segmentation Models

The segmentation phase includes pixelwise localization of stroke lesions to create lesion probability maps that will later be utilized as spatial priors for segmentation-guided classification. The development of four different segmentation architectures was completed using only stroke-positive images (Ischemic and Bleeding) and their associated ground truth masks.

Training configuration:

- **Optimizer:** Adam
- **Learning rate:** 1×10^{-4}
- **Batch size:** 32 **Epochs:** 30
- **Loss:** Hybrid BCE–Dice loss

The hybrid loss function is defined as follows:

$$L_{\text{seg}} = \lambda_1 L_{\text{BCE}} + \lambda_2 L_{\text{Dice}} \quad (3)$$

In Equation (3),

- L_{seg}: total segmentation loss.
- λ_1, λ_2: weighting coefficients balancing the two loss terms.

1) Baseline Segmentation Models: There are three baseline models: U-Net, ResUNet, Attention U-Net. Both U-Net [13] and ResUNet are encoder-decoder neural networks with skip connections that allow information preserved in the encoder to be used in the decoder; they do this through deep supervision where intermediate decoder outputs have additional individual segmentation heads and provide additional gradient signal to the model to help it learn to segment ischemic lesions that have indistinct margins. ResUNet's encoder [14] is based on a pretrained ResNet34 architecture making it a residual neural network (provide residual connections to improve gradient flow) and leveraging features from ImageNet for improved stroke tissue contrast discrimination on CT scans. Attention U-Net [15] improves U-Net's skip connections by adding scSE (spatial and channel Squeeze-and-Excitation) attention blocks [6] to the decoder which recalibrates the feature at both the spatial and channel levels to suppress background regions/tissues (ex. skull) and enhance stroke lesion regions. Therefore, all three architectures have shown adequate performance when used to segment and provide a strong basis for the evaluation of other segmentation architectures.

2) Swin-UNet: Swin-UNet is selected as one of the segmentation models due to its ability to model long-range spatial dependencies and capture global contextual information. Swin-Unet utilizes a pre-trained Swin Transformer Encoder [16] with shifted window self-attention; therefore, it is highly efficient at modeling global context while retaining pixel resolution (local context). Its unique CNN decoder produces pixel-level lesion maps by reconstructing the image. Swin-Unet is structured as a multi-task architecture to simultaneously produce both classification output using a global bottleneck head and generate a segmentation map from the decoder, with only the segmentation output being used in the downstream segmentation-guided classification stage.

The encoder uses shifted window self-attention mechanism [16] which is defined as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \quad (4)$$

In Equation (4),

- Q, K, V: query, key, and value matrices derived from input tokens
- d: dimensionality used for scaling

The shifted window methodology allows a certain amount of interaction between neighboring windows, thereby enhancing continuity of feature representations across spatially separate defects. This is highly relevant to the medical imaging space, since many defect boundaries will span multiple spatial regions.

The segmentation output of Swin-Unet is subsequently post-processed using thresholding to create binary masks of the lesions. A threshold of 0.85 was used with Swin-Unet for segmentation while 0.25 was used with the CNN-based segmentation models, which highlight the relative differences in output distributions' degree of sharpness between Swin-Transformer and CNN decoders. In summary, Swin-Unet model's superior ability to dominate both global and local features allows it to yield an improved Dice score [18] and IoU when compared to the other segmentation models.

D. Model Selection and Segmentation-Guided Classification

To ensure architectural diversity for the ensemble, a structured model selection procedure is applied after all models are trained independently. The ten classifiers are divided into five groups based on their architectural lineage:

- **Group A** (Dense connectivity): DenseNet121-SE, DenseNet201-SE
- **Group B** (Efficient/Lightweight): EfficientNet-B4, EfficientNetV2-S, MobileNetV2
- **Group C** (Residual): ResNet50, ResNet101
- **Group D** (Inception-style): InceptionV3, Xception
- **Group E** (Modern CNN): ConvNeXt-Small

The representative model for each group is the one achieving the highest macro-recall on the Validation set. Macro-recall is used as the selection criterion because it equally penalizes false negatives across all classes of detection, an important consideration, owing to the potential consequences of incorrectly classifying ischemic stroke. For the four segmentation models evaluated, the model which achieves the best Dice Score [18] on the validation set is the representative for segmentation.

The five selected classifiers are: DenseNet201-SE (Group A), MobileNetV2 (Group B), ResNet50 (Group C), Xception (Group D), and ConvNeXt-Small (Group E). Swin-UNet is selected as the segmentation representative.

The five selected classifiers are then fine-tuned using an attention mechanism driven by segmentation, where the lesion probability map from the frozen segmentation model, Swin-UNet, is employed as a spatial constraint during the feature extraction process.

Training configuration (Segmentation-guided Classification):

- **Optimizer:** AdamW **Weight decay:** 1×10^{-4}
- **Learning rate:** 1×10^{-4}
- **Batch size:** 32 **Epochs:** 10
- **Loss:** Weighted cross-entropy with label smoothing ($\epsilon = 0.1$)

Let $F \in \mathbb{R}^{H \times W \times C}$ represent the intermediate feature maps from the classification backbone, and $M \in \mathbb{R}^{1 \times H' \times W'}$ the lesion probability map. The map is resized to match the spatial dimensions of F. A convolutional attention gate G is computed as:

$$G = \sigma(\text{BN}(\text{Conv}([F,\ M_{\text{scaled}}]))) \quad (5)$$

In Equation (5),

- $[F,\ M_{\text{scaled}}]$: channel-wise concatenation of the feature map and resized segmentation map.
- $\text{Conv}(\cdot)$: 1×1 convolution reducing concatenated channels back to C.
- $\text{BN}(\cdot)$: batch normalization for training stability.
- $\sigma(\cdot)$: sigmoid producing spatial gate $G \in [0, 1]^{H \times W \times C}$.

The gated feature map with amplified lesion-relevant regions and suppressed background is given as follows:

$$F' = F \odot G \tag{6}$$

In Equation (6),

- F: original feature map of the backbone.
- G: learned spatial attention gate conditioned on the segmentation prior.

The workflow diagram for the development of the confidence-weighted ensemble model using the segmentation-guided method is illustrated in Fig. 2.

The gate G is fully learned during fine-tuning, and is particularly beneficial for detection of ischemic stroke, where lesion boundaries are often indistinct.

E. Proposed Confidence Weighted Ensemble Model

Predictions from the five segmentation-guided classifiers are combined using the **Confidence Weighted Ensemble**. All five members receive both the input image and the segmentation priors during inference.

Training configuration:

- **Optimizer:** Adam
- **Learning rate:** 1×10^{-4}
- **Batch size:** 32 **Epochs:** 10
- **Loss:** Weighted negative log-likelihood (NLL) loss
- **Backbone parameters:** Frozen; only fusion weights and temperatures are trained

The calibrated probability for each member is computed using temperature-scaled softmax:

$$P_i = \text{softmax}\left(\frac{z_i}{T_i}\right) \tag{7}$$

In Equation (7),

- z_i: output logits from the i-th segmentation-guided classifier.
- T_i: per-model temperature; higher T_i produces softer probability distributions.

The final ensemble prediction is as follows:

$$P_{final} = \sum_{i=1}^{n} w_i \, P_i, \qquad w_i = \frac{\exp(\ell_i)}{\sum_{j=1}^{n} \exp(\ell_j)} \tag{8}$$

In Equation (8),

- $n = 5$: number of ensemble members.
- ℓ_i: learnable scalar logit weight for the i-th member.

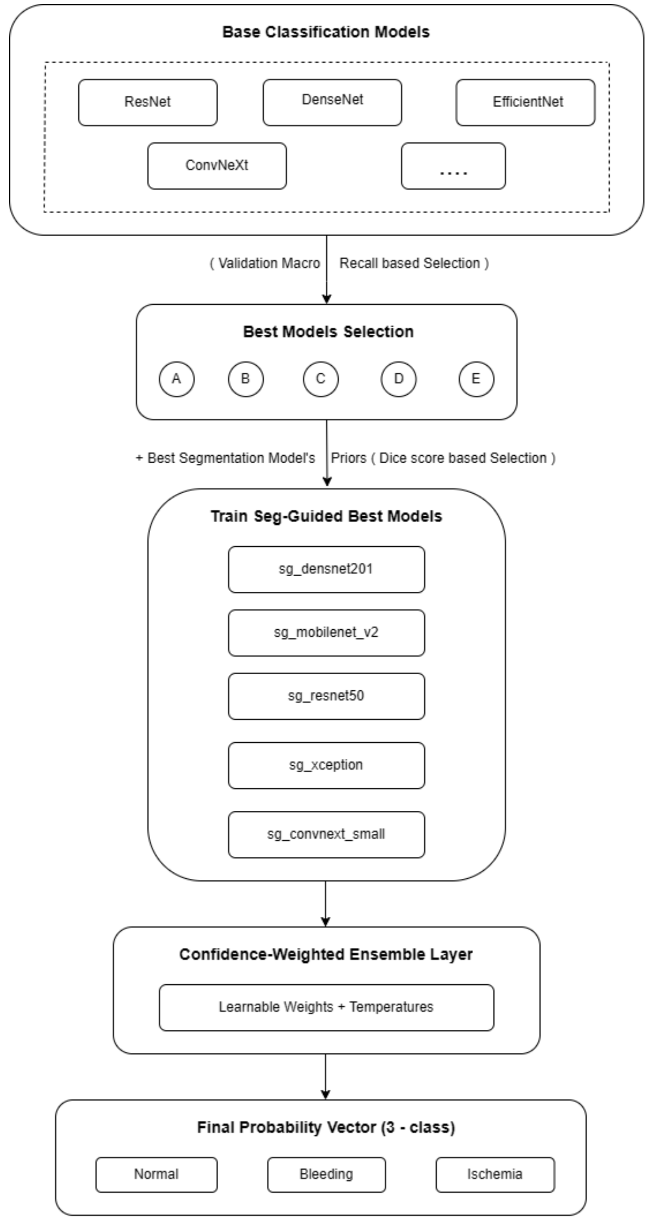

Fig. 2. Confidence Weighted Ensemble Model Pipeline

- w_i: normalized fusion weight derived by softmax over ℓ_i.
- T_i: learnable temperature (shared with Equation (7)).

Both ℓ_i and T_i are jointly optimized during ensemble training. Freezing of the backbone parameters will create a stable environment for fusion training, while maintaining the integrity of the previously trained representations. The results show that this ensemble approach reduces the incidence of missed detections and misclassifications between Normal and Ischemic classes which are the most frequently confused.

F. Explainability

Grad-CAM++ effectively aids the generation of class-discriminative heat maps that visualize spatial areas [17] determining predictions produced by the model. The generated heatmap is overlaid on the source CT scan file(s) and constrained to brain tissue using a morphologically derived brain mask. The proportion of the affected region is calculated as the number of lesion pixels that were activated to the total number of pixels within the brain; therefore providing both a quantitative measure of how extensive the lesion is, as well as a visual explanation for the prediction produced by the model.

III. Results

This section presents a comprehensive evaluation of the proposed framework on the held-out test set. Section III-A reports the performance of all ten classification models across accuracy, Macro AUC, Macro F1-score, Macro Precision, and Macro Recall, providing the basis for group representative selection. All classification metrics are reported as percentages.

Section III-B evaluates the four segmentation models using Dice score [18], Intersection over Union (IoU), and Hausdorff distance [19], where Dice measures region overlap, IoU captures the proportion of the union covered by the prediction, and Hausdorff distance quantifies worst-case boundary deviation in millimeters.

Section III-C presents the quantitative performance of the proposed confidence-weighted ensemble along with a comparative discussion highlighting the benefits of diversity-aware fusion and segmentation guidance.

Section III-D provides segmentation outputs for stroke classes and Grad-CAM++ explainability visualizations for all three diagnostic classes.

A. Classification Performance

Table I presents the performance of all ten classification models. Models marked with † are those selected as group representatives for segmentation-guided fine-tuning and ensemble construction. The **Recall** column reports macro-recall on *Test* set.

TABLE I
Performance comparison of classification models. Values in %.

Model	Precision	Recall	AUC	Accuracy	F1
ConvNeXt-Small†	99.00	98.80	99.90	99.10	98.90
InceptionV3	98.59	98.63	99.69	98.74	98.61
Xception†	98.42	97.94	99.67	98.47	98.17
MobileNetV2†	98.25	98.14	99.49	98.43	98.19
ResNet50†	97.94	97.87	99.52	98.16	97.90
ResNet101	97.60	97.79	99.47	97.89	97.69
DenseNet121-SE	96.06	96.04	99.64	96.54	96.05
DenseNet201-SE†	96.37	95.71	99.51	96.54	96.02
EfficientNetV2-S	94.71	93.96	99.37	94.92	94.32
EfficientNet-B4	93.88	91.26	98.69	93.30	92.40

ConvNeXt-Smal achieved the best overall performance, demonstrating strong capability in capturing both local lesion characteristics and global anatomical context through large-kernel convolutions and layer normalization. InceptionV3, Xception, and MobileNetV2 also performed competitively, benefiting from efficient multi-scale and depthwise feature extraction. ResNet variants showed stable performance through residual learning, with ResNet50 selected as one of the segmentation-guided models due to its slightly better validation recall. DenseNet models remained competitive through feature reuse, whereas the EfficientNet variants exhibited comparatively lower performance, suggesting greater sensitivity to dataset-specific tuning requirements.

B. Segmentation Performance

Table II reports segmentation performance on stroke-positive test images. The model marked with † is selected as the segmentation representative for segmentation-guided classification training.

TABLE II
Performance comparison of Segmentation Models

Model	Dice	IoU	Hausdorff (mm)
Swin-UNet†	**0.7533**	**0.6497**	**11.33**
Attention U-Net	0.7012	0.5892	28.99
U-Net	0.6425	0.5143	30.26
ResUNet	0.6424	0.5069	49.64

The Swin-UNet outperformed all methods in terms of performance and accuracy, with its superior boundary accuracy attributed to its ability to capture long distance connections through the use of transformers. Attention U-Net achieved second highest performance due its improved ability to learn lesion-focused features when using scSE attention blocks instead of a basic U-Net. Although both U-Net and ResUNet achieved similar Dice scores, the higher Hausdorff distance [19] for ResUNet indicated a tendency towards boundary outliers and over-segmentation. Due to its superior performance and boundary accuracy, Swin-UNet will be used for all subsequent stages.

C. Ensemble Performance

Table III presents the performance of the proposed Confidence Weighted Ensemble on the held-out test set.

TABLE III
Performance of the Confidence Weighted Ensemble

Metric	Value (%)
Accuracy	99.19
Macro F1-Score	99.04
AUC	99.96

The Confidence Weighted Ensemble achieved superior performance compared to the best individual classifier (ConvNeXt-Small), demonstrating the advantages of fusing classifiers with the use of both diversity and segmentation guidance. Furthermore, the ensemble provided the most substantial reductions in misclassification rates between the normal and ischemic classes (normal and ischemic strokes) which were the most frequently confused classes by the individual models.

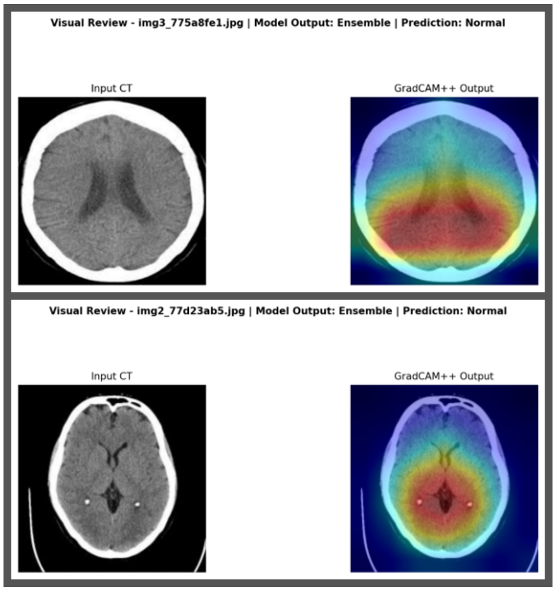

Fig. 3. GradCAM++ Outputs for Normal brain cases

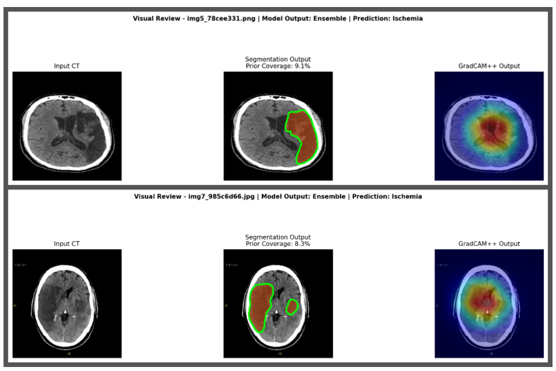

Fig. 4. Segmentation, GradCAM++ Outputs for Ischemic stroke cases

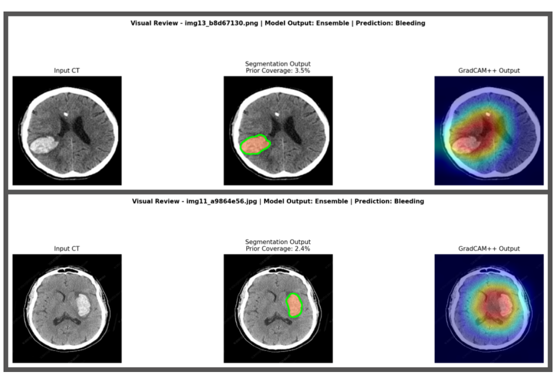

Fig. 5. Segmentation, GradCAM++ Outputs for Bleeding stroke cases

D. *Qualitative Results and Explainability Visualizations*

Figures 3, 4 and 5 present qualitative outputs of the proposed framework across all three diagnostic classes. Each example includes the input CT image, the corresponding segmentation output for abnormal cases, and the Grad-CAM+ heatmap highlighting the spatial regions most influential to the ensemble prediction.

IV. Conclusion

This paper presents an end-to-end deep learning framework for the automated classification and segmentation of stroke from CT brain images. The classification portion of the framework consists of ten independent classification models that have been trained in isolation to generate multiple unique feature representations; the segmentation portion consists of four independent segmentation models that also have been trained in isolation for the same purpose. A structured process was then used to select the highest macro recall representative classification model from each of five architecture families, and then fine-tuned those classifier models using a learned spatial attention gating mechanism based on the lesion probability maps resulting from the best-performing segmentation model, Swin-UNet [16].

A **Confidence Weighted Ensemble** of the five segmentation-guided classifiers, with jointly optimized fusion weights and per-model temperature parameters, achieves a classification accuracy of 99.19% and AUC of 99.96%, outperforming the best individual model. Swin-UNet achieves superior segmentation performance with a Dice score of 0.7533 and Hausdorff distance [19] of 11.33 mm. With Grad-CAM++, class-discriminative heat maps [17] are provided so that qualitative visual prediction explanations, along with quantified lesion size estimates, enhance the arrangement in the clinical interpretation of the prediction. It demonstrates that it is a comprehensive and interpretable method for an automated diagnosis of stroke via the use of computed tomography (CT) imaging.

V. Future Scope

Despite the high effectiveness of the proposed approach, several directions can broaden its utility in clinical practice. Combining CT and MRI would provide additional tissue contrast to improve diagnostic accuracy. Extending the framework from 2D slice analysis to full 3D volumetric processing would enable richer spatial context. In addition, if compressed model methods (e.g., pruning, quantization, knowledge distillation) could be used, they could facilitate delivery of the method in time-sensitive and emergency situations. Bayesian approaches to uncertainty estimation would allow the system to identify low-confidence predictions so that they could be presented to an expert for review. Federated learning approaches would allow multiple sites to train the same model without requiring patient data be transferred from one site to another, providing large-scale validation necessary for clinical viability.

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